

BCA SEMESTER V • INTRODUCTION TO MACHINE LEARNING • UNIT IV

Multivariate Methods: Data, Parameter Estimation & Missing Values

Foundational statistical parameters and handling incomplete observations.



Prescribed Textbook: Ethem Alpaydin, *Introduction to Machine Learning* (2004), Chapter 5

Unit IV Roadmap & Scope

Today we focus strictly on the foundational three topics from Alpaydin Chapter 5:



1. Multivariate Data

Vectors, matrices, and multi-feature observations.



2. Parameter Estimation

Mean vectors, variances, and covariance matrices.



3. Missing Values

Imputation strategies using means and relationships.

What is Multivariate Data?

Variable (Feature)

A measurable characteristic or attribute (e.g., Height, Weight).

Observation (Sample)

A single distinct entity or instance being measured.

Multivariate Data

Multiple variables recorded simultaneously for each observation.

OBSERVATION AS A COLUMN VECTOR

$$\mathbf{x} = x_1, x_2, \dots, x_d^T$$

- $\mathbf{x}$: An observation vector
- x_j : Measurement of the j -th variable
- d : Dimensionality (total number of features)

From One Variable to Multiple Variables

UNIT III Univariate Data

Only a single variable is observed per entity.

Classroom Example: Measuring only student Height: $x = 165$ cm

Analyzed using scalar mean and variance.

UNIT IV Multivariate Data

Multiple variables are recorded simultaneously.

Classroom Example: Measuring Height, Weight, and Age together: $x = [165, 60, 20]^T$

Captures dependencies between attributes.

The Data Matrix • Structure & Notation

Sample (t)	x ₁ (Height)	x ₂ (Weight)	x ₃ (Age)
Sample 1 (x ^{t=1})	170	65	21
Sample 2 (x ^{t=2})	160	55	20
Sample 3 (x ^{t=3})	175	72	22

Simple Classroom Example Dataset (N = 3, d = 3)

Rows = Observations (Samples)

There are **N** observations, indexed by $t = 1, \dots, N$.

Columns = Variables (Features)

There are **d** variables, indexed by $j = 1, \dots, d$.

Notation: x_j^t is variable j of sample t .

The Mean Vector (μ or m)

In multivariate data, each variable has its own mean. Combining them yields the **mean vector**:

$$\mathbf{m} = m_1, m_2, \dots, m_d^T$$

Estimator for each component:

$$m_j = \frac{\sum_{t=1}^N x_j^t}{N}$$

Simply sum each column and divide by the number of samples N .

Simple Classroom Example

Observations for (x_1, x_2) :

- Sample 1: $[2, 4]^T$
- Sample 2: $[4, 8]^T$

$$m_1 = (2 + 4) / 2 = 3$$

$$m_2 = (4 + 8) / 2 = 6$$

$$\mathbf{m} = [3, 6]^T$$

Variance in Multivariate Data

Definition of Variance

Measures how widely individual values of a single feature spread around their respective mean.

Individual Variances

Each of the d features possesses its own sample variance: $s_1^2, s_2^2, \dots, s_d^2$.

SAMPLE VARIANCE FORMULA

$$s_j^2 = \frac{\sum_{t=1}^N x_j^t - m_j^2}{N}$$

Simple Classroom Example (x_1): values = 2, 4; mean = 3

$$s_1^2 = [(2 - 3)^2 + (4 - 3)^2] / 2 = (1 + 1) / 2 = 1.0$$

Covariance Between Two Variables

Covariance tells us how two different variables vary or change together.



Positive Covariance

Both variables increase or decrease together.



Negative Covariance

One variable increases as the other decreases.



Zero Covariance

No linear relationship between the two features.

The Covariance Matrix (Σ or S)

$$S = \begin{bmatrix} s_1^2 & s_{12} & \dots & s_{1d} \\ s_{21} & s_2^2 & \dots & s_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ s_{d1} & s_{d2} & \dots & s_d^2 \end{bmatrix}$$

A square $d \times d$ symmetric matrix.

Diagonal Elements ($s_{jj} = s_j^2$)

Represent individual feature variances.

Off-Diagonal Elements (s_{ij})

Represent pairwise covariances ($s_{ij} = s_{ji}$).

Key Property: S is always symmetric because $\text{Cov}(x_i, x_j) = \text{Cov}(x_j, x_i)$.

What is Parameter Estimation?

"Parameter estimation means using observed sample data to estimate the unknown parameters of the underlying distribution or model."

— Ethem Alpaydin, Chapter 5

UNIT III: UNIVARIATE

Estimated scalar parameters (θ): single mean μ and single variance σ^2 .



UNIT IV: MULTIVARIATE

Estimates vector of means (m) and matrix of covariances (S).

The Two Primary Parameters

Multivariate data is primarily summarized by location and dispersion parameters:



1. Mean Vector ($\mathbf{m}$)

Location parameter representing the central point.

Contains $\mathbf{d}$ individual parameters.



2. Covariance Matrix ($\mathbf{S}$)

Dispersion parameter describing spread and interactions.

Contains $\mathbf{d(d + 1) / 2}$ unique values.

Parameter Estimation: Simple Numerical Example

CLASSROOM DATASET (N=3, D=2)

Sample	x_1	x_2
$t = 1$	2	4
$t = 2$	4	6
$t = 3$	6	8

ESTIMATION STEPS

Step 1 (Mean of x_1): $m_1 = (2 + 4 + 6) / 3 = 4$

Step 2 (Mean of x_2): $m_2 = (4 + 6 + 8) / 3 = 6$

Resulting Mean Vector: $m = [4, 6]^T$

Variances and covariances are similarly estimated from the deviations from these means.

The Problem of Missing Values

INCOMPLETE DATASET EXAMPLE

Sample	Feature 1	Feature 2	Feature 3
Sample 1	10	20	30
Sample 2	12	?	32
Sample 3	14	24	34

What to do when a value is missing?

Sensor failure, patient non-response, or unrecorded entries.

Why not just discard the row?

Discarding Sample 2 wastes valid measurements in Feature 1 and Feature 3.

What is Missing Value Estimation?

"Missing value estimation, or imputation, means estimating a missing value using the available information."

— Ethem Alpaydin, Chapter 5

Primary Objective

Complete the dataset so standard machine learning algorithms can run without dropping rows.

Information Preserved

Retains the observation's remaining known attributes for accurate model training.

Mean Imputation

CLASSROOM CALCULATION EXAMPLE

Known values for a feature: 10, 12, ?, 14

Mean of available observations:

$$(10 + 12 + 14) / 3 = 12$$

Replace ? with 12.

Advantage

Extremely simple and leaves the overall sample mean unchanged.

Limitation

Artificially reduces variance and ignores relationships with other variables.

Estimation Using Other Variables

Instead of using an isolated mean, we leverage the correlations among attributes:

Predictive Concept (Alpaydin Ch 5)

Treat the missing variable as a target output, and the known variables as inputs.

Preserves relationships and dependencies between features.

Simple Classroom Example

Known: Study Hours = 6, Past Marks = 80



Estimate: Missing Final Mark $\approx$ 82

Unit IV Core Summary

Multivariate Data	Multiple variables measured simultaneously for each observation vector.
Mean Vector ($\mathbf{m}$)	Vector collecting the independent arithmetic mean of every feature.
Variance & Covariance	Variance measures individual spread; covariance measures joint variation.
Covariance Matrix ($\mathbf{S}$)	Square symmetric matrix with variances on diagonal and covariances off-diagonal.
Missing Value Imputation	Completing records via mean imputation or inter-variable predictions.

Exam-Oriented Questions (Alpaydin Ch 5)

Short-Answer Questions

- Define multivariate data and observation vector.
- What is a mean vector? State its formula.
- Define covariance and its physical meaning.
- Explain diagonal vs off-diagonal elements in S .
- What is imputation in missing value estimation?

Long-Answer Questions

- Explain multivariate data representation with an example.
- Describe parameter estimation for multivariate data.
- Discuss mean vector and covariance matrix properties.
- Compare mean imputation with relationship-based estimation.